# MANCHESTER TECHNOLOGIES COLLEGE

## GRAND TEST – JEE

**TOTAL MARKS – 300 MARKS** **DURATION – 3.00 HOURS** **DATE – 16-07-2026**

### PART A - PHYSICS

**Q1** The displacement of a charge Q in the electric field $E = e_1\,\hat{\imath} + e_2\,\hat{\jmath} + e_3\,\hat{k}$ is $r = a\,\hat{\imath} + b\,\hat{\jmath}$. The work done is

**(A)** $Q(e_1 a + e_2 b)$ **(B)** $Q\sqrt{(e_1 a)^2 + (e_2 b)^2}$

**(C)** $Q(e_1 + e_2)\sqrt{a^2 + b^2}$ **(D)** $Q\left(\sqrt{e_1{}^2 - e_2{}^2}\right)(a + b)$

**Q2** Given that a photon of light of wavelength 10,000 Å has an energy equal to 1.23 eV. When light of wavelength 5000 Å and intensity $I_0$ falls on a photoelectric cell, the saturation current is $0.40 \times 10^{-6}$ A and the stopping potential is 1.36 V. If the cathode and the anode are kept at the same potential, the emitted electrons:

**(A)** All have the same KE equal to 1.36 eV **(B)** All have the average KE equal to $\frac{1.36}{2}\ eV$

**(C)** All have the maximum KE equal to 1.36 eV **(D)** All have the minimum KE equal to 1.36 eV

**Q3** The nuclear radius of a nucleus with nucleon number 16 is $3 \times 10^{-15}$ m. Then, the nuclear radius of a nucleus with nucleon number 128 is

**(A)** $3 \times 10^{-15}$ m **(B)** $1.5 \times 10^{-15}$ m **(C)** $6 \times 10^{-15}$ m **(D)** $4.5 \times 10^{-15}$ m

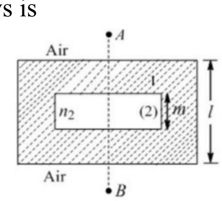
**Q4** The angular momentum of an electron in hydrogen atom is $4h/2\pi$. Kinetic energy of this electron is

**(A)** 4.35 eV **(B)** 1.51 eV **(C)** 0.85 eV **(D)** 13.6 eV

**Q5** In a Young's double-slit experiment, 12 fringes are observed to be formed in a certain segment of the screen when light of wavelength 600 nm is used. If the wavelength of light is changed to 400 nm, the number of fringes observed in the same segment of the screen is given by

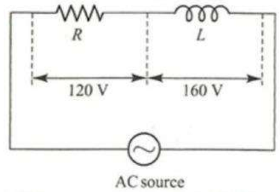
**(A)** 12 **(B)** 18 **(C)** 24 **(D)** 30

**Q6** In a thick glass slab of thickness l, and refractive index $n_1$, a cuboidal cavity of thickness 'm' is carved as shown in the figure and is filled with a liquid of R.I. $n_2$ $(n_1 > n_2)$. The ratio l/m, so that shift produced by this slab is zero when an observer A observes an object B with paraxial rays is




**(A)** $\frac{n_1 - n_2}{n_2 - 1}$ **(B)** $\frac{n_1 - n_2}{n_2(n_1 - 1)}$ **(C)** $\frac{n_1 - n_2}{n_1 - 1}$ **(D)** $\frac{n_1 - n_2}{n_1(n_2 - 1)}$

**Q7** The circuit given in the figure has a resistance less choke coil L and a resistance R. The voltages across R and L are 120 V and 160 V respectively as shown. The rms value of the applied AC source voltage is

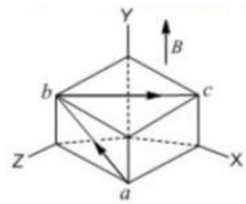



**(A)** 100 V **(B)** 200 V **(C)** 300 V **(D)** 400 V

**Q8** A horizontal ring of radius r = 1/2 m is kept in a vertical constant magnetic field of 1 T. The ring is collapsed from maximum area to zero area in 1 s. Then the e.m.f. induced in the ring is

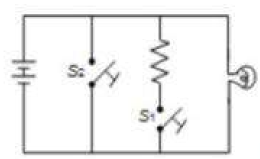
**(A)** 1 V **(B)** $\left(\frac{\pi}{4}\right)V$ **(C)** $\left(\frac{\pi}{2}\right)V$ **(D)** $\pi\ V$

**Q9** Two straight segments of wire ab and bc, each carrying current I, are placed as shown in the figure. The cube edge is 50 cm and the magnetic field is uniform along the Y-axis having magnitude 0.4 T. If I = 3 A, the force experienced by wire abc in the presence of this magnetic field is




**(A)** $0.6\,\hat{\imath}$ **(B)** $1.2(\hat{\imath} + \hat{k})$ **(C)** $0.6(\sqrt{2}\,\hat{\imath} + \hat{\jmath} - \sqrt{2}\,\hat{k})$ **(D)** $0.6(\sqrt{2}\,\hat{\imath} - \hat{k})$

**Q10** Which of the two switches, $S_1$ and $S_2$, shown in the figure will produce short-circuiting?




**(A)** $S_1$ **(B)** $S_2$ **(C)** Both $S_1$ and $S_2$ **(D)** Neither $S_1$ nor $S_2$

**Q11** The quantity X = $\frac{\varepsilon_0 LV}{t}$, where $\varepsilon_0$ is the permittivity of free space, L is length, V is potential difference and t is time. The dimensions of X are same as that of

**(A)** Resistance **(B)** Charge **(C)** Voltage **(D)** Current

**Q12** A solid sphere of radius R has a charge Q distributed in its volume with a charge density $\rho = kr^a$, where k and a are constants and r is the distance from its centre. If the electric field at r = R/2 is 1/8 times that at r = R, find the value of a.

**(A)** 2 **(B)** 3 **(C)** 2.5 **(D)** 0.2

**Q13** Two tuning forks A and B give 4 beats/s when sounded together. The frequency of A is 320 Hz. When some wax is added to B and it is sounded with A, 4 beats/s per second are again heard. The frequency of B is

**(A)** 312 Hz **(B)** 316 Hz **(C)** 324 Hz **(D)** 328 Hz

**Q14** A body of mass m is released from a height h onto a scale pan hung from a spring. The spring constant of the spring is k, the mass of the scale pan is negligible and the body does not bounce relative to the pan; then the amplitude of vibration is

**(A)** $\frac{mg}{k}\sqrt{1 - \frac{2hk}{mg}}$ **(B)** $\frac{mg}{k}$ **(C)** $\frac{mg}{k} + \frac{mg}{k}\sqrt{1 + \frac{2hk}{mg}}$ **(D)** $\frac{mg}{k} - \frac{mg}{k}\sqrt{1 - \frac{2h}{mg}}$

**Q15** Two moles of ideal helium gas are in a rubber balloon at 30°C. The balloon is fully expandable and can be assumed to require no energy for its expansion. The temperature of the gas in the balloon is slowly changed to 35°C. The amount of heat required in raising the temperature is nearly (take R = 8.31 J/mol.K)

**(A)** 62 J **(B)** 104 J **(C)** 124 J **(D)** 208 J

**Q16** If g is the acceleration due to gravity on the earth's surface, the gain in the potential energy of an object of mass m raised from the surface of the earth to a height equal to the radius R of the earth is

**(A)** ½ $mgR$ **(B)** 2mgR **(C)** mgR **(D)** ¼ $mgR$

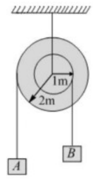
**Q17** In the pulley system shown, if the radii of the bigger and smaller pulley are 2 m and 1 m respectively, and the acceleration of block A is 5 m/s² in the downward direction, the acceleration of block B will be

**(A)** $0\frac{m}{s^2}$

**(B)** $5\frac{m}{s^2}$

**(C)** $10\frac{m}{s^2}$

**(D)** $\frac{5}{2}\frac{m}{s^2}$




**Q18** The potential energy of a particle is determined by the expression $U = \alpha(x^2 + y^2)$, where α is a positive constant. The particle begins to move from a point with coordinates (3, 3), only under the action of the potential field force. Then its kinetic energy T at the instant when the particle is at a point with the coordinates (1, 1) is

**(A)** $8\alpha$ **(B)** $24\alpha$ **(C)** $16\alpha$ **(D)** Zero

**Q19** A block is lying on a horizontal frictionless surface. One end of a uniform rope is fixed to the block, which is pulled in the horizontal direction by applying a force F at the other end. If the mass of the rope is half the mass of the block, the tension in the middle of the rope will be

**(A)** F **(B)** $\frac{2F}{3}$ **(C)** $\frac{3F}{5}$ **(D)** $\frac{5F}{6}$

**Q20** A body is projected horizontally from the top of a tower with initial velocity 18 m/s. It hits the ground at an angle of 45° to the horizontal. What is the vertical component of velocity when it strikes the ground?

**(A)** $9\, ms^{-1}$ **(B)** $9\sqrt{2}\, ms^{-1}$ **(C)** $18\, ms^{-1}$ **(D)** $18\sqrt{2}\, ms^{-1}$

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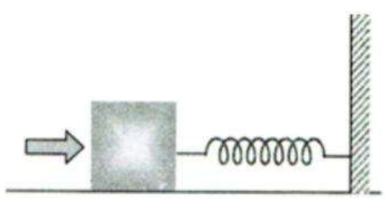
## Section - B (Numerical Value Questions)

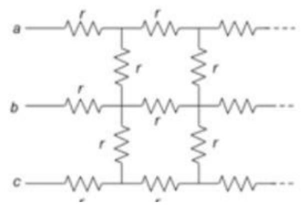
**Q21** A block of mass 0.18 kg is attached to a spring of force-constant 2 N/m. The coefficient of friction between the block and the floor is 0.1. Initially the block is at rest and the spring is un-stretched. An impulse is given to the block as shown in the figure. The block slides a distance of 0.06 m and comes to rest for the first time. The initial velocity of the block in m/s is V = N/10. Then N is



**Q22** A frog sits on the end of a long board of length L = 5 m. The board rests on a frictionless horizontal table. The frog wants to jump to the opposite end of the board. What is the minimum take-off speed (in m/s), i.e., relative to ground 'v', that allows the frog to do the trick? The board and the frog have equal masses.

**Q23** A solid sphere of radius R has a charge Q distributed in its volume with a charge density $\rho = kr^a$, where k and a are constants and r is the distance from its centre. If the electric field at r = R/2 is 1/8 times that at r = R, find the value of a.

**Q24** In the infinite grid shown, if the value of each resistance is r = 2(√5 − 1) Ω. Find the equivalent resistance between the points a and c (in Ω).




**Q25** A Bohr hydrogen atom undergoes a transition n = 5 to n = 4 and emits a photon of frequency f. The frequency of circular motion of the electron in the n = 4 orbit is $f_4$. The ratio f/$f_4$ is found to be 18/5m. State the value of m.

## PART B - CHEMISTRY

### Section - A (Multiple Choice Questions)

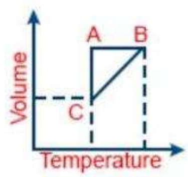
**Q26** The number of molecules in 200 mL of each of $O_2$, $NH_3$ and $CO_2$ at STP is

**(A)** In the order $CO_2 < O_2 < NH_3$ **(B)** In the order $NH_3 < O_2 < CO_2$ **(C)** $NH_3 = CO_2 < O_2$ **(D)** Same

**Q27** Which of the following pairs has both members from the same group of the periodic table?

**(A)** $Mg - Ba$ **(B)** $Mg - Na$ **(C)** $Mg - Cu$ **(D)** $Mg - K$

**Q28** Five moles of a gas is put through a series of changes as shown graphically in a cyclic process (Volume vs Temperature graph with points A, B and C). The processes A→B, B→C and C→A respectively are




**(A)** Isochoric, Isobaric, Isothermal **(B)** Isobaric, Isochoric, Isothermal
**(C)** Isothermal, Isobaric, Isochoric **(D)** Isochoric, Isothermal, Isobaric

**Q29** For the reaction, $2NO_2(g) \rightleftharpoons 2NO(g) + O_2(g)$ when Kp and Kc are compared at 184°C it is found that

**(A)** whether Kp is greater than, less than or equal to Kc depends upon the total gas pressure
**(B)** $Kp = Kc$
**(C)** Kp is less than Kc
**(D)** Kp is greater than Kc

**Q30** The charge/size ratio of a cation determines its polarizing power. Which one of the following sequences represents the increasing order of the polarizing power of the cationic species $K^+$, $Ca^{2+}$, $Mg^{2+}$, $Be^{2+}$?

**(A)** $Mg^{2+}$, $Be^{2+}$, $K^+$, $Ca^{2+}$ **(B)** $Be^{2+}$, $K^+$, $Ca^{2+}$, $Mg^{2+}$ **(C)** $K^+$, $Ca^{2+}$, $Mg^{2+}$, $Be^{2+}$ **(D)** $Ca^{2+}$, $Mg^{2+}$, $Be^{2+}$, $K^+$

**Q31** Ga is below Al in the periodic table, but the atomic radius of Ga is less than that of Al. It is because of

**(A)** lanthanoid contraction **(B)** greater screening effect **(C)** inert pair effect **(D)** none of these

**Q32** Arrange in order of decreasing trend towards $S_E$ (electrophilic substitution) reactions: (I) chlorobenzene (II) benzene (III) anilinium chloride (IV) toluene

**(A)** IV > II > I > III **(B)** I > II > III > IV **(C)** II > I > III > IV **(D)** III > I > II > IV

**Q33** Match List I with List II:

| | **List I Element detected** | | **List II Reagent used/ Product formed** |
|---|---|---|---|
| A | Nitrogen | I | $Na_2[Fe(CN)_5NO]$ |
| B | Sulphur | II | $AgNO_3$ |
| C | Phosphorous | III | $Fe_4[Fe(CN)_6]_3$ |
| D | Halogen | IV | $(NH_4)_2MoO_4$ |

**(A)** A-II, B-IV, C-I, D-III **(B)** A-IV, B-II, C-I, D-III **(C)** A-II, B-I, C-IV, D-III **(D)** A-III, B-I, C-IV, D-II

**Q34** Match List I with List II:

| | LIST-I Name of reaction | | LIST-II Reagent used |
|---|---|---|---|
| A | Hell-Volhard-Zelinsky reaction | I | $NaOH + I_2$ |
| B | Iodoform reaction | II | (i) $CrO_2Cl_2$, $CS_2$ (ii) $H_2O$ |
| C | Etard reaction | III | (i) $Br_2$/ red phosphorus (ii) $H_2O$ |
| D | Gatterman-Koch reaction | IV | CO, HCl, anhyd. $AlCl_3$ |

**(A)** A-III, B-II, C-I, D-IV **(B)** A-III, B-I, C-IV, D-II **(C)** A-I, B-II, C-III, D-IV **(D)** A-III, B-I, C-II, D-IV

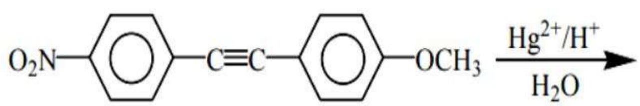
**Q35** 2 mole of an ideal gas at 27°C expands isothermally and reversibly from a volume of 4 litre to 40 litre. The work done (in kJ) by the gas is:

**(A)** w = −28.72 kJ **(B)** w = −11.488 kJ **(C)** w = −5.736 kJ **(D)** w = −4.988 kJ

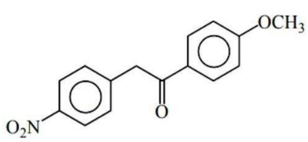
**Q36** In an organic compound of molar mass 108 g mol$^{-1}$, C, H and N atoms are present in 9 : 1 : 3.5 by mass. The molecular formula can be:

**(A)** $C_6H_8N_2$ **(B)** $C_7H_{10}N$ **(C)** $C_5H_6N_3$ **(D)** $C_4H_{18}N_3$

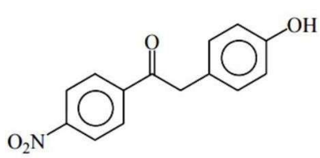
**Q37** The major product obtained from the reaction is:



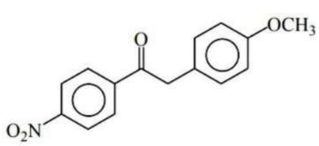

**(A)**



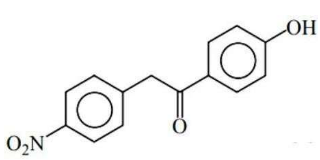

**(B)**




**(C)**



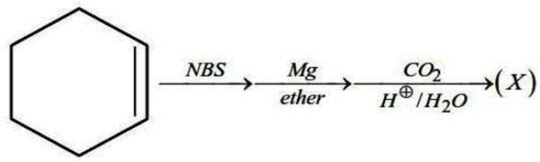

**(D)**

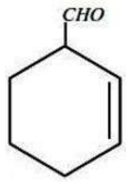
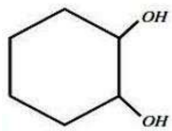


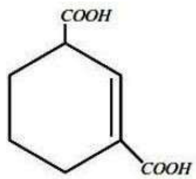
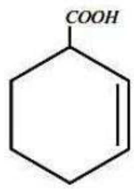

**Q38** If x is the specific resistance (in S$^{-1}$cm) of the electrolyte solution and y is the molarity of the solution, then $\Lambda_m$ (in S cm$^2$ mol$^{-1}$) is given by:

**(A)** $\frac{1000x}{y}$ **(B)** $1000\ \frac{y}{x}$ **(C)** $\frac{1000}{xy}$ **(D)** $\frac{xy}{1000}$

**Q39** In the given reaction: (X) will be:



**(A)**  **(B)** 

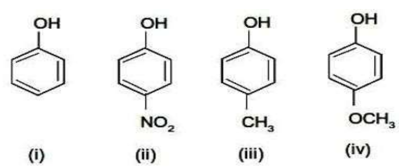
**(C)**  **(D)** 

**Q40** Assertion: $[Ni(CO)_4]$ is diamagnetic in nature.
Reason: Ni atom undergoes $dsp^2$ hybridisation.

**(A)** Both assertion and reason are true, and reason is the correct explanation of assertion **(B)** Both assertion and reason are false
**(C)** Assertion is false, reason is true **(D)** Assertion is true, but reason is false

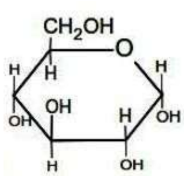
**Q41** Which among the following named reactions results in a decrease in the number of carbons in the carbon chain?
**(A)** Aldol condensation **(B)** Wurtz reaction **(C)** Hoffmann bromamide degradation **(D)** Fittig reaction

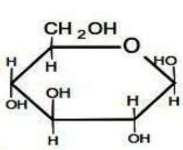
**Q42** Given four compounds: (i) phenol, (ii) p-nitrophenol, (iii) p-cresol (p-methylphenol), (iv) p-methoxyphenol. The correct order of acidity is

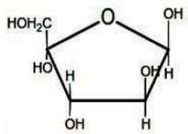


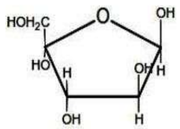
**(A)** ii > iv > i > iii **(B)** ii > i > iii > iv **(C)** iv > ii > i > iii **(D)** ii > iii > i > iv

**Q43** The structure of β-D-glucopyranose is:

**(A)** 

**(B)** 

**(C)** 

**(D)** 

**Q44** Number of unpaired electrons in the $B_2$ molecule is

**(A)** 2 **(B)** 6 **(C)** 8 **(D)** 10

**Q45** Choose the correct statement:

**(A)** $CCl_4$ is non-polar because C–Cl bond is non-polar **(B)** $CCl_4$ is polar because C–Cl bond is polar

**(C)** Though C–Cl bond is polar, $CCl_4$ is a non-polar molecule **(D)** Though C–Cl bond is non-polar, $CCl_4$ is a polar molecule

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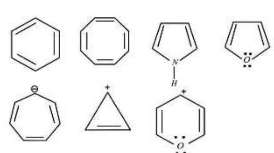
## Section - B (Numerical Value Questions)

**Q46** In infinite dilution, the equivalent conductance's of $Ba^{2+}$ and $Cl^{-}$ are 127 and 76 $ohm^{-1}cm^{-1}$ eqvt$^{-1}$. The equivalent conductivity of $BaCl_2$ at indefinite dilution is [Nearest integer]

**Q47** The wavelength of an electron of kinetic energy $4.50 \times 10^{-29}$ J is ...... $\times 10^{-5}$ m. (Nearest integer). Given: mass of electron is $9 \times 10^{-31}$ kg, h = $6.6 \times 10^{-34}$ J s

**Q48** Find out the number of aromatic compounds or ions from the following



**Q49** How many of the following compounds are polar? $CH3\ Cl, NH3, CO2\ , H2O, CH4, BF3$

**Q50** How many unpaired electrons are present in the central atom of $[Fe(ox)_3]^{3-}$?

## PART C - MATHEMATICS

## Section - A (Multiple Choice Questions)

**Q51** Let A = {x ∈ R : |x + 1| < 2} and B = {x ∈ R : |x − 1| ≥ 2}. Then which one of the following statements is NOT true?
**(A)** $A - B = (-1, 1)$ **(B)** $B - A = R - (-3, 1)$ **(C)** $A \cap B = (-3, -1]$ **(D)** $A \cup B = R - [1, 3)$

**Q52** Let $f(n) = \left[\frac{1}{3} + \frac{3n}{100}\right] n$, where [n] denotes the greatest integer less than or equal to n. Then $\sum_{n=1}^{56} f(n)$ is equal to:
**(A)** 56 **(B)** 689 **(C)** 1287 **(D)** 1399

**Q53** The number of solutions of the equation $\cos\left(x + \frac{\pi}{3}\right)\cos\left(\frac{\pi}{3} - x\right) = \frac{1}{4}\cos^2 2x \in [-3\pi, 3\pi]$ is:
**(A)** 8 **(B)** 5 **(C)** 6 **(D)** 7

**Q54** If non-zero real numbers b and c are such that min f(x) > max g(x), where $f(x) = x^2 + 2bx + 2c^2$ and $g(x) = -x^2 - 2cx + b^2, (x \in R)$; then |c/b| lies in the interval:
**(A)** $\left(0, \frac{1}{2}\right)$ **(B)** $\left(\frac{1}{2}, \frac{1}{\sqrt{2}}\right)$ **(C)** $\left(\frac{1}{\sqrt{2}}, \sqrt{2}\right)$ **(D)** $(\sqrt{2}, \infty)$

**Q55** If the domain of the function $f(x) = log_e(4x^2 + 11x + 6) + sin^{-1}(4x + 3) + cos^{-1}\left(\frac{10x+6}{3}\right)$], then 36|α + β| is equal to:
**(A)** 63 **(B)** 45 **(C)** 72 **(D)** 54

**Q56** The sum $\sum_{r=1}^{10}(r^2 + 1) \times (r!)$ is equal to:
**(A)** $(11!)$ **(B)** $10 \times (11!)$ **(C)** $101 \times (11!)$ **(D)** $11 \times (11!)$

**Q57** In the expansion of $\left(\frac{x}{cos\theta} + \frac{1}{x\, sin\theta}\right)^{16}$ $l_1$ is the least value of the term independent of x when π/8 ≤ θ ≤ π/4, and $l_2$ is the least value of the term independent of x when π/16 ≤ θ ≤ π/8, then the ratio $l_2 : l_1$ is equal to:
**(A)** 16 : 1 **(B)** 8 : 1 **(C)** 1 : 8 **(D)** 1 : 16

**Q58** Let $S_n = 1 \cdot (n-1) + 2 \cdot (n-2) + 3 \cdot (n-3) + \ldots + (n-1) \cdot 1, n \geq 4$. The sum $\sum_{n=4}^{\infty}\left(\frac{2S_n}{n!} - \frac{1}{(n-2)!}\right)$ is equal to:
**(A)** $\frac{e-1}{3}$ **(B)** $\frac{e-2}{6}$ **(C)** $\frac{e}{3}$ **(D)** $\frac{e}{6}$

**Q59** The equation of the straight line passing through the point of intersection of the family of lines x(3 + 4λ) + y(4 + 3λ) + 1 + 6λ = 0, which is at a minimum distance from the point (1, 1), can be expressed as x + by − c = 0, where b and c are natural numbers. Then the value of (b + c) is
**(A)** 5 **(B)** 8 **(C)** 9 **(D)** 10

**Q60** If a tangent to the circle $x^2 + y^2 = 1$ intersects the coordinate axes at distinct points P and Q, then the locus of the midpoint of PQ is
**(A)** $x^2 + y^2 - 2xy = 0$ **(B)** $x^2 + y^2 - 16x^2y^2 = 0$ **(C)** $x^2 + y^2 - 4x^2y^2 = 0$ **(D)** $x^2 + y^2 - 2x^2y^2 = 0$

**Q61** A line makes the same angle θ with each of the x-axis and z-axis. If the angle β which it makes with the y-axis is such that $sin^2\beta = 3sin^2\theta\, cos^2\theta$ **(A)** $\frac{2}{3}$ **(B)** $\frac{1}{5}$ **(C)** $\frac{3}{5}$ **(D)** $\frac{2}{5}$

**Q62** The value of $\cot\left[\sum_{n=1}^{23} cot^{-1}\left(1 + \sum_{k=1}^{n} 2k\right)\right]$
**(A)** $\frac{23}{25}$ **(B)** $\frac{25}{23}$ **(C)** $\frac{23}{24}$ **(D)** $\frac{24}{23}$

**Q63** Let A = $\begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}$ (a 2×2 matrix with rows [0, −2] and [2, 0]). If M and N are two matrices given by $M = \sum_{k=1}^{10} A^{2k}$ and $N = \sum_{k=1}^{10} A^{2k-1}$ then $MN^2$ is

**(A)** a non-identity symmetric matrix **(B)** a skew-symmetric matrix **(C)** neither symmetric nor skew-symmetric matrix **(D)** an identity matrix

**Q64** The number of values of k for which the system of linear equations, (k+2)x + 10y = k, kx + (k+3)y = k − 1 has no solution, is:

**(A)** 1 **(B)** 2 **(C)** 3 **(D)** infinitely many

**Q65** Let f : [1, 3] → R be a function satisfying x/[x] ≤ f(x) ≤ √(6−x), for all x ≠ 2, and f(2) = 1, where R is the set of all real numbers and [x] denotes the largest integer less than or equal to x.

Statement 1: $\underset{x\to 2}{f(x)}$ exists.

Statement 2: f is continuous at x = 2.

**(A)** Statement 1 is true, Statement 2 is true, Statement 2 is a correct explanation for Statement 1 **(B)** Statement 1 is false, Statement 2 is true **(C)** Statement 1 is true, Statement 2 is true, Statement 2 is not a correct explanation for Statement 1 **(D)** Statement 1 is true, Statement 2 is false

**Q66** Let S = {(λ, μ) ∈ R × R : f(t) = (|λ|$e^{|t|}$ − μ)·sin(2|t|), t ∈ R is a differentiable function}. Then S is a subset of:

**(A)** $R \times [0,\infty)$ **(B)** $[0,\infty) \times R$ **(C)** $R \times (-\infty,0)$ **(D)** $(-\infty,0) \times R$

**Q67** The set of all real values of λ for which the function $f(x) = (1 - cosx)^2(\lambda + sinx)$, x ∈ (−π/2, π/2), has exactly one maxima and exactly one minima, is:

**(A)** $\left(-\frac{1}{2},\frac{1}{2}\right) - \{0\}$ **(B)** $\left(\frac{1}{2},\frac{1}{2}\right)$ **(C)** $\left(\frac{3}{2},\frac{3}{2}\right)$ **(D)** $\left(-\frac{3}{2},\frac{3}{2}\right) - \{0\}$

**Q68** The area of the region $\{(x,y): x^2 \le y \le 8 - x^2, y \le 7\}$ is

**(A)** 21 **(B)** 18 **(C)** 24 **(D)** 20

**Q69** If the components of $\vec{a}$ along and perpendicular to $\vec{b} = 3\hat{\imath} + \hat{\jmath} - \hat{k}$ respectively are $\frac{16}{11}\left(3\hat{\imath} + \hat{\jmath} - \hat{k}\right)$ and $\frac{1}{11}\left(-4\hat{\imath} - 5\hat{\jmath} - 17\hat{k}\right)$, then $\alpha^2 + \beta^2 + \gamma^2$ is equal to (where $\vec{a} = \alpha\hat{\imath} + \beta\hat{\jmath} + \gamma\hat{k}$):

**(A)** 23 **(B)** 18 **(C)** 16 **(D)** 26

**Q70** If the shortest distance between the lines $L_1 : \vec{r} = (2 + \lambda)\hat{\imath} + (1 - 3\lambda)\hat{\jmath} + (3 + 4\lambda)\hat{k}, \lambda \in R$ and $L_2 : \vec{r} = 2(1 + \mu)\hat{\imath} + 3(1 + \mu)\hat{\jmath} + (5 + \mu)\hat{k}, \mu \in R$ is $\frac{m}{\sqrt{n}}$, where gcd(m, n) = 1, then the value of m + n equals.

**(A)** 384 **(B)** 387 **(C)** 377 **(D)** 390

## Section - B (Numerical Value Questions)

**Q71** Let f(x) be continuous at x = 0. If f″(0) exists and $\frac{2f(x) - 3a\ \ (2x) + bf(8x)}{\underset{x\to 0}{sin^2x}}$ists, with f(0) ≠ 0, f′(0) ≠ 0, then the value of 3a/b is

**Q72** Let the mean and variance of four numbers 3, 7, x and y (x > y) be 5 and 10 respectively. Then the mean of the four numbers 3 + 2x, 7 + 2y, x + y and x − y is _______.

**Q73** Let {x} and [x] denote the fractional part and the greatest integer respectively of a real number x. If $\int_0^n \{x\}dx, \int_0^n [x]dx$ and $10^{\frac{n^2-n}{2}}$ (n ∈ N, n > 1) are three consecutive terms of a G.P., then n is equal to_____

**Q74** Mr. A has two fair cubic dice: one with faces numbered from 2 to 7 and the second with faces numbered from 4 to 9. Twice, he randomly picks one of the dice (each die is equally likely) and rolls. If it is known that the sum of the resulting two rolls is 10, then the probability he rolled the same die twice is m/√n, gcd(m, n) = 1. Find the least value of (n − m).

**Q75** Let y = y(x) be the solution of the differential equation $\frac{dy}{dx} = \frac{4y^3+2yx^2}{3xy^2+x^3}$, y(1) = 1. If y(2) ∈ [n−1, n) for some n ∈ N, then n is equal to ______